

AYMANE AIT DADS

AI & Data Science Analyst Intern | LLM Experimentation · Python · Data-Driven Business Insights

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PROFESSIONAL SUMMARY

Engineering student at EURECOM with hands-on LLM fine-tuning experience, currently ranking top 10% (0.80/1.0) on the NVIDIA Nemotron Kaggle competition (\$106K+ prize pool) by training a 30B Mamba-Transformer MoE model with LoRA adapters and conducting 20+ single-variable ablation studies. At Orange Maroc, built Random Forest and K-Means models on 100K+ network records, reducing manual analysis time by ~60% and delivering Power BI dashboards with executive maintenance recommendations. Proficient in Python, Hugging Face Transformers, Claude API, LLM orchestration, RAG, and prompt engineering - all required to build and compare AI models against critical business goals. Available for Summer 2026; holds a French student visa and is open to USA/Canada placement pending sponsorship.

CORE COMPETENCIES

LLM Experimentation | Python Proficiency | AI Model Comparison | Data Preparation | Prompt Engineering | Data Storytelling | Experiment Documentation | Actionable Insights

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Prepared and engineered features from **100K+ fiber-optic network records** (upload speed, latency, jitter) enabling Random Forest and K-Means models to classify quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, output adopted directly by the network optimization team for capacity planning.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows built entirely in Python with Pandas and Scikit-learn.
- Delivered **Power BI dashboards** and stakeholder presentations translating model outputs into data-driven maintenance recommendations for executive and technical audiences.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score (0.80/1.0)** in a \$106K+ prize competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought logic examples using SFTTrainer.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict sequence - including OOM monkey-patches, Mamba kernel stubs, and Triton ptxas fixes - and documented 20+ single-variable ablation experiments (packing, loss masking, adapter size) to identify the optimal training configuration.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · TRL · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · 1st in Class

- Ranked **1st on the EURECOM private class leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual learning rates on 250K images spanning 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark, applying 10-view TTA (flips, crops, blur, resizes) and a 3-model ensemble weighted by validation AUC to maximize out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle T4 GPU

Twitter Sentiment Analysis

EURECOM NLP Course · 2025 · End-to-End Business NLP Pipeline

- Built an **end-to-end NLP pipeline** progressing from tokenization and TF-IDF through word2vec embeddings to transformer fine-tuning, benchmarking each stage for sentiment classification accuracy.
- Conducted **systematic hyperparameter optimization** across model variants, documenting experiment results and translating findings into actionable recommendations aligned with data storytelling best practices.

Python · Hugging Face Transformers · Scikit-learn · Pandas · word2vec

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, NLP, Computer Vision, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Power BI | Scikit-learn | Feature Engineering
AI & LLM Experimentation: Hugging Face Transformers | LoRA / QLoRA | Prompt Engineering | Claude API | RAG | LangChain | vLLM
MLOps & Visualization: PyTorch | Git | Linux | FP16 Mixed Precision | Ablation Studies | Matplotlib | Seaborn